

ME-255 A:FLUID MECHANICS I (M)

For

**B.E II YEAR IV SEMESTER
(MECHANICAL ENGINEERING)**



By

Dr. S.K.SINGH

Professor

**Department of Mechanical Engineering,
M.B.M. Engineering College,
J.N.V. University, Jodhpur (Rajasthan)**

(UNIT -5)

DIMENSIONAL ANALYSIS

LECTURE 28

Dimensional analysis

Introduction Dimensional analysis deals with a process in which all important variables involved in a physical phenomenon are systematically organised into dimensionless groups.

Uses

- 1) Testing dimensional homogeneity of any equation of fluid motion.
- 2) Deriving equations expressed in terms of non-dimensional parameters to show relative significance of each parameter.
- 3) Planning model tests and presenting experimental results in terms of non-dimensional parameters.

Dimensions

All physical quantities are expressed by magnitude and units.

In SI system FUNDAMENTAL or PRIMARY PHYSICAL QUANTITIES which are assigned base units are.

- ✓ 1) Mass
- ✓ 2) Length
- ✓ 3) Time
- ✓ 4) Temperature
- 5) Current
- 6) Luminous Intensity

In fluid mechanics and heat transfer first four are useful and given dimensions as

M, L, T, θ respectively.

DERIVED or SECONDARY QUANTITIES

They are expressed in terms of fundamental quantities.

Note: Sometimes Force (F) is used as fundamental quantity instead of mass. $F-L-T-\theta$ System.

② Dimensions of derivatives and Integral

$$i) \frac{dy}{dx} \Rightarrow \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta y}{\Delta x} \right) = \frac{y}{x}$$

$$ii) \frac{d^2y}{dx^2} \Rightarrow \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d \left(\frac{dy}{dx} \right)}{\Delta x \rightarrow 0} = \frac{\frac{dy}{dx}}{x} = \frac{y/x}{x} = \frac{y}{x^2}$$

$$iii) \int y dx = y \cdot x$$

$$iv) \iint p \cdot dx dy = p \cdot x \cdot y$$

Dimensions of some important physical quantities

- 1) Angular speed (Rad./sec.) = T^{-1}
- 2) Angular acceleration (Rad./sec.²) = T^{-2}
- 3) Angular momentum (Mxrxv) = $M L^2 T^{-1}$
- 4) moment or torque (Fxr) = $M L^2 T^{-2}$
- 5) Stream function, ψ ($u = \frac{\partial \psi}{\partial y}$) = $L^2 T^{-1}$
- 6) velocity potential, ϕ ($u = \frac{\partial \phi}{\partial x}$) = $L^2 T^{-1}$
- 7) Circulation, Γ ($\Gamma = \oint \vec{v} \cdot d\vec{s}$) = T^{-1}
- 8) Vorticity, $\left(\frac{\Gamma}{\Delta A} \right)$ = T^{-1}
- 9) Rotation, $\frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$ = T^{-1}
- 10) Surface tension (F/l) = $M T^{-2}$
- 11) Dynamic viscosity (μ) ($\tau = \mu \frac{du}{dy}$) = $M L^{-1} T^{-1}$
- 12) Kinematic viscosity (ν) = μ/ρ = $L^2 T^{-1}$
- 13) modulus of elasticity (E or k) = $M L^{-1} T^{-2}$

Note! Rotation = $\frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) = T^{-1}$

Circulation = $\Gamma = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) dx dy = L^2 T^{-1}$

Vorticity = $\frac{\text{Circulation}}{\text{Area}} = T^{-1}$

Dimensional Homogeneity

Principle of dimensional homogeneity states that "Equations describing a physical phenomenon must be dimensionally homogeneous i.e. the dimensions of various terms on two sides of equations are identical."

example

The equation (Euler's equation of motion)

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} \quad \text{is dimensionally correct.}$$

$$[L T^{-1}] \left[\frac{L T^{-1}}{L} \right] + [L T^{-1}] \left[\frac{L T^{-1}}{L} \right] = [M^{-1} L^3] \left[\frac{M L^{-1} T^{-2}}{L} \right]$$

$$[L T^{-2}] + [L T^{-2}] = [L T^{-2}]$$

dimensions of both sides are same.

Note: There are several empirical relations in which some coefficient is there. If coefficient is assumed dimensionless then the equation is dimensionally non-homogeneous

e.g. Chezy's formula

$$V = C \sqrt{m i}$$

V = velocity through a channel
 m = Hydraulic mean depth or hydraulic radius
 i = slope of channel
 C = Chezy's coefficient.

$$L T^{-1} = 1 [L^{1/2}] [1]$$

(dimensions of both sides are not same)

Hence if equation has to be dimensionally homogeneous then Chezy's coefficient (C) must have dimensions

$$C = \frac{L T^{-1}}{L^{1/2}} = L^{1/2} T^{-1}$$

$$C = L^{1/2} T^{-1}$$

Also Manning coefficient (n) in the Manning formula for velocity of flow in open channel

$$V = \frac{1}{n} m^{2/3} i^{1/4}$$

$$n = L^{1/3} T$$

LECTURE 29

(4)

Applications of dimensional homogeneity

Principle of dimensional homogeneity serves the following purpose.

- 1) Helps to check whether an equation of any physical phenomenon is dimensionally homogeneous or not.
- 2) It helps to determine the dimensions of a physical quantity.

eg. $q = -KA \frac{dT}{dx}$, we can find the dimensions of thermal conductivity K .

- 3) Dimensional homogeneity helps to convert the units from one system to another.

Dimensional analysis: Buckingham's $\Pi(C\pi)$ theorem

"If there are n variables in a dimensionally homogeneous equation and if these variables contain m primary dimensions. Then these variables can be grouped in $(n-m)$ independent non-dimensional parameters and called $\Pi(C\pi)$ terms."

Note: There can be formed in finite number of dimensionless groups but only $(n-m)$ will be independent.

Procedure for Buckingham method

- ① List all important variables involved in a physical phenomenon.
- ② Note the primary dimensions involved in them and find out the value of m . Then there will be $(n-m)$ non-dimensional groups.
- ③ Now we recognise m repeating variables which are common to all groups desired to be formed. They are so chosen that all m fundamental quantities are included.
- ④ Non repeating variables are taken one by one together with repeating variables.
- ⑤ In fluid flow problem, generally characteristic length (l), characteristic velocity (V) and fluid density (ρ) are taken as repeating variables. The reason is that l , V and ρ are representative of inertial behaviour of the fluid. They include all fundamental dimensions.
 $l [L]$, $V [LT^{-1}]$, $\rho [ML^{-3}]$

Guide lines for recognition of important variables in a problem

- These variables are most important physical quantities in the problem.
 - Non repeating independent variables should be listed as far as possible.
 - These should be geometric (e.g. length), Kinematic (e.g. velocity) and dynamic (e.g. force) variables in the list.
 - Relevant fluid property (e.g. ρ or μ) must be included.
- For heat transfer problem, a thermal property (e.g. thermal conductivity, k) are also chosen.

⑥

- ⊗ None of repeating variable should be dimensionless.
- ⊗ No two variables have same dimensions.
- ⊗ They themselves do not form a dimensionless parameter.
- ⊗ None of repeating variable should be dependent.

⑥ π terms are formed by m repeating variables and one non repeating variable.

eg. $\left\{ \begin{array}{l} \text{Viscosity } (\mu) \\ \text{Gravitational acceleration } (g) \\ \text{Surface tension } (\sigma) \end{array} \right\}$
non repeating variables.

⑦ Write expressions for π terms. Raise unknown exponent to repeating variable and put exponent (1) on non-repeating variable.

⑧ Find values of unknown exponents by use of fundamental dimensions.

Some facts about π terms

In order to get expression in desired form, the following points are noted.

- ① If a quantity is dimensionless, it is a π term.
- ② If two quantities have same dimensions, their ratio will be a π term e.g. (H/d) .
- ③ Any π term may be replaced by any power of that term
eg. $\pi_1 \rightarrow \pi_1^{-1}$, $\pi_2 \rightarrow \pi_2^2$, $\pi_3 \rightarrow \frac{1}{\sqrt{\pi_3}}$
- ④ Any π term may be replaced by multiplying it by a numerical constant. eg. $\pi_1 \rightarrow 2\pi_1$
- ⑤ Any π term may be replaced by multiplying or dividing it by another π term
eg. $\pi_1 \text{ or } \pi_2 \rightarrow (\pi_1 \times \pi_2) \text{ or } (\pi_1 / \pi_2)$

LECTURE 30

Important force ratios: Dimensionless numbers

(7)

(i) Reynolds Number (Re)

It is the ratio of inertia force to the viscous force.

$$Re = \frac{\text{Inertia force}}{\text{Viscous force.}}$$

$$\begin{aligned} \text{Inertia force} &= \text{mass} \times \text{acceleration} = (\rho \times \text{Volume}) \times \frac{\text{Velocity}}{\text{time}} \\ &= \rho \times A \times \text{length} \times \frac{\text{Velocity}}{\text{time}} \end{aligned}$$

$$\text{Inertia force} = \rho l^2 v^2$$

$$\text{Viscous force} = \tau \times A = \mu \frac{v}{l} l^2 = \mu v l$$

$$\tau = \mu \frac{du}{dy}$$

$$Re = \frac{\rho l v^2}{\mu v} = \frac{\rho v l}{\mu}$$

$$Re = \frac{\rho v l}{\mu}$$

l = characteristic length
e.g. diameter of pipe.

Reynolds number will be considered for situations in which viscous force predominates over other forces e.g. gravity force and elastic forces.
e.g. in following situations.

(i) Completely enclosed flow

- ① - incompressible flow through pipes, Venturimeters, orifice meters and flow between parallel plates.

(ii) Completely submerged flow

- ② - Low velocity flow around aeroplanes and submarines
- ③ - Flow around structures and bodies.
- ④ - Flow through low speed aerodynamic machines.

⑧

(ii) Froude number (Fr)

$$Fr = \left(\frac{\text{Inertia force}}{\text{Gravity force}} \right)^{1/2}$$

$$\text{Gravity force} = mg = \rho l^3 g$$

$$Fr = \frac{\sqrt{\rho l^2 v^2}}{\sqrt{\rho l^3 g}} = \frac{v}{\sqrt{lg}}$$

Froude number is used for dynamic similarity when the forces other than gravitational forces are negligible e.g. in

Note - It is the ratio of mean velocity of ship to velocity of small wave in quiet fluid.
- Its value is considered as critical value.

- (1) Flow in open channels (free surface flows)
- (2) Flow of liquid jets from orifice
- (3) Surface wave motion created by boats & ships.
- (4) Flow over notches, weirs etc.
- (5) problems in which fluid of different mass densities flow over one another.

(iii) Euler Number (Eu)

It is the ratio of pressure force to the inertia force.

$$Eu = \frac{\text{Pressure force}}{\text{Inertia force}} = \frac{\rho \Delta^2}{\rho l^2 v^2}$$

$$\text{Pressure force} = \text{Pressure} \times \Delta^2 = p \Delta^2$$

$$E = \frac{p}{\rho v^2}$$

↓
It is also called pressure coefficient.

Euler number is used when

pressure changes in a flow are predominant.

- (1) Flow through pipes in which turbulence is fully developed and viscous forces are negligible.

- (2) Flow through pen stock leading water to hydraulic turbine.
- (3) Flow over submerged bodies.
- (4) Pressure rise due to sudden closure of valves.
- (5) Also used where phenomenon of cavitation takes place.

(iv) Mach Number (M) $M = \left(\frac{\text{Inertia force}}{\text{Elastic force}} \right)^{1/2}$

Elastic force = bulk modulus of elasticity $\times \text{Area} = K l^2$
 $= \rho c^2 l^2$

$$c = \sqrt{K/\rho}$$

$$\Rightarrow K = \rho c^2$$

c = Speed of sound in flowing medium

$$M = \left(\frac{\rho l^2 v^2}{\rho l^2 c^2} \right)^{1/2}$$

$$M = v/c = \frac{\text{Speed of fluid}}{\text{Speed of sound in that fluid}}$$

Mach number effects are considered only when M exceeds 0.4 and effects of compressibility comes into picture e.g. in

- (i) Aerodynamic testing.
- (ii) Flow of aeroplanes and projectiles through air at supersonic speed.
- (iii) In case unsteady flows with water hammer problems.
- (iv) Flow of gases exceeding velocity of sound.

Transonic flow	$0.9 < M < 1.1$
Supersonic flow	$1 < M < 7$
Hypersonic flow	$M > 7$

(v) Weber Number (W)

$$W = \frac{\text{Inertia force}}{\text{Surface tension force}}$$

$$= \frac{\rho l^2 v^2}{\sigma \times l} = \frac{\rho l v^2}{\sigma}$$

$$\sigma = \frac{\text{Surface tension}}{\text{per unit length}}$$

Weber number is important for following cases.

- (1) Capillary tube flows (Capillary rise in narrow passages, capillary movement of water in soil)
- (2) Human blood flow studies in veins and arteries.
- (3) Thin sheet flows (very thin sheet of fluid flowing over a surface).

(10)

Model investigation

Engineers associated with design and construction of various hydraulic structures e.g. dams and spillways or hydraulic machines e.g. pumps and turbines, first perform experiments on the models and find out how structure would behave when it's actually constructed.

- A MODEL is a small scale replica of the actual structure or machine.
- A PROTOTYPE is a full scale structure employed in actual engg. design.

It is often desirable and advantageous to go for model studies of full prototype because information resulting from model studies may indicate certain potential problems which can be rectified before prototype's built model studies involve.

- (i) model design and fabrication
- (ii) performance test on models and analysis of test results.

Similitude Results obtained from experiments on models can be applied to prototype only if there exists a complete similarity between model and prototype. For complete similarity following type of similarity must exist between model and prototype.

① Geometric similarity

Geometric similarity exists between model and prototype if ratio of corresponding geometric dimensions in them are equal. Geometric similar systems may differ in size but identical in shape.

For geometric similarity

$$\text{Length scale ratio } (L_r) = \frac{L_m}{L_p} = \frac{b_m}{b_p} = \frac{d_m}{d_p} = \text{Scale ratio}$$

length → breadth → depth →

$$\text{Area scale ratio} = A_r = \frac{A_m}{A_p} = \frac{L_m \times b_m}{L_p \times b_p} = \frac{(\text{Scale ratio})^2}{L_r^2} \quad (11)$$

$$\text{Volume scale ratio} = V_r = \frac{V_m}{V_p} = \frac{L_m \times b_m \times d_m}{L_p \times b_p \times d_p} = \frac{(\text{Scale ratio})^3}{L_r^3}$$

(b) Kinematic Similarity

Kinematic similarity exists between model and prototype if (i) The paths of homologous moving particles are geometrically similar.

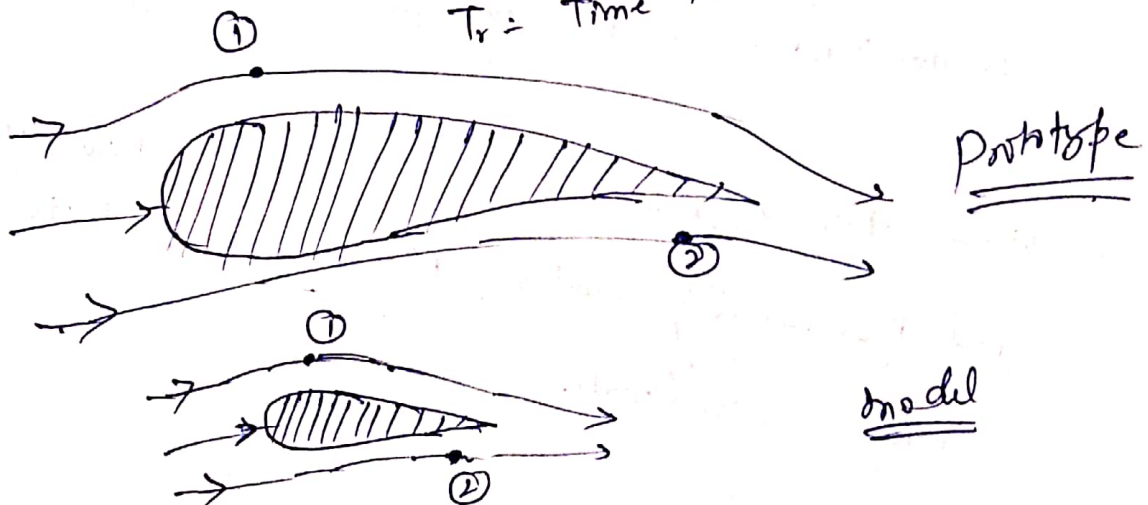
(ii) The ratio of kinematic parameters (velocity and acceleration) of homologous particles are equal. Since they have magnitude and direction. Hence direction of velocity and acceleration in two flows are same, and also velocity at corresponding points have constant ratio.

$$\text{velocity scale ratio} = V_r = \frac{V_m}{V_p}$$

Geometric similarity is prerequisite for kinematic similarity

$$\text{Thus } \frac{V_m}{V_p} = \frac{L_m / T_m}{L_p / T_p} = \frac{L_m}{L_p} \times \frac{T_p}{T_m} = \frac{L_r}{T_r}$$

$$L_r = \text{Length scale ratio} = \frac{T_m}{T_p}$$



$$\frac{(V_1)_m}{(V_1)_p} = \frac{(V_2)_m}{(V_2)_p} = V_r$$

Point 1 and 2 represents homologous points in model and prototype

(12)

acceleration scale ratio = $a_r = \frac{a_m}{a_p}$

$$\text{discharge ratio} = Q_r = \frac{Q_m}{Q_p} = \frac{A_m}{A_p} \times \frac{V_m}{V_p} = (L_r)^2 \times V_r$$

© Dynamic Similarity

Dynamic similarity exists between model and prototype if the ratio of all the forces acting at homologous points in the two systems are equal.

Both geometric and kinematic similarity are prerequisite for dynamic similarity.

For dynamic similarity dimensionless force ratios should be equal for the two systems

e.g. $Re = \frac{\text{Inertia force}}{\text{Viscous force}} \Rightarrow \boxed{Re_m = Re_p}$

- In actual flow situations, it may be impossible to satisfy dynamic similarity with respect to all the forces. Also some forces may be predominant while others are insignificant. Two systems are then treated to be dynamic similar if the ratio of inertia to the predominant force is same for the given systems.

- When two systems are geometrically, kinematically and dynamically similar, they are said to have complete similitude.

Types of models

models can be classified under two categories.

(i) Undistorted models

An undistorted model is geometrically similar to its prototype. i.e. scale ratio of corresponding linear dimensions are same. In this case, results of model can be directly applied to its prototype.

(ii) Distorted models

In distorted models geometric scale ratio is different in different directions. Such models with distortion of dimensions is frequently adopted in river models, where a different scale ratio for depths is adopted.

Note If we keep the same scale ratio for depth of river model, depth of flow would be too small (0.6-0.7 cm).

- (i) Surface tension effects would be there and flow situation will alter. (ii) Flow in the model may become laminar corresponding to turbulent flow in prototype.
- (iii) Precise measurement would be difficult for small depths.
- (iv) Current and eddies in small depths would not be strong enough to transport the sediment.

Hence vertical scale is enhanced resulting in a distorted model.

merits and demerits of distorted model

MERITS

- (i) Precise measurements are possible due to increase in depth of fluid.

(14)

- (ii) The effect of surface tension can be reduced to minimum.
- (iii) model size can be sufficiently reduced resulting in cost saving.
- (iv) Tractive force can be created to bring about a satisfactory movement of silt and sand of the model.

DEMERITS

- (i) Depth distortion give rise to different wave patterns.
- (ii) Pressure and velocity distribution are not truly reproduced due to different horizontal and vertical scales.
- (iii) Results obtained from distorted models are difficult to extrapolate and interpolate.

Scale effect

If complete similarity does not exist between model and prototype. There will be discrepancy between results obtained from model tests and those which will be indicated by prototype after its construction.

This discrepancy is called scale effect.

TUTORIAL 10

Tutorial problem (Solved)

Consider force F on the propeller of an aircraft. It depends on forward speed of aircraft U , density of air ρ , viscosity of air μ , diameter of propeller D and speed of rotation of propeller N . $F = f[U, \rho, \mu, D, N]$
Obtain expression for F in non-dimensional terms.

Solution:

total variables, $n = 6$

Total primary dimensions, $m = 3$, using Buckingham
theorem number of non-dimensional π terms = $n - m = 6 - 3 = 3$

choose repeating variables, D, U, ρ

π_1 will have

π_2 "

π_3 "

$$\left. \begin{array}{l} D U \rho F \\ D U \rho \mu \\ D U \rho N \end{array} \right\} \Rightarrow \begin{array}{l} \pi_1 = D^{a_1} U^{b_1} \rho^{c_1} F \\ \pi_2 = D^{a_2} U^{b_2} \rho^{c_2} \mu \\ \pi_3 = D^{a_3} U^{b_3} \rho^{c_3} N \end{array}$$

① π_1

$$M^0 L^0 T^0 = [L]^{a_1} [L T^{-1}]^{b_1} [M L^{-3}]^{c_1} [M L T^{-2}]$$

$$M^0 L^0 T^0 = [M]^{c_1+1} [L]^{a_1+b_1-3c_1+1} [T]^{-b_1-2}$$

$$c_1+1=0 \Rightarrow c_1=-1, \quad -b_1-2=0 \Rightarrow b_1=-2, \quad a_1+b_1-3c_1+1=0$$

$$a_1-2+3+1=0 \Rightarrow a_1=-2$$

$$\pi_1 = \frac{F}{\rho U^2 D^2}$$

② π_2

$$M^0 L^0 T^0 = [L]^{a_2} [L T^{-1}]^{b_2} [M L^{-3}]^{c_2} [M L^{-1} T^{-1}]$$

$$= [M]^{c_2+1} [L]^{a_2+b_2-3c_2+1} [T]^{-b_2-1}$$

$$c_2+1=0 \Rightarrow c_2=-1, \quad -b_2-1=0 \Rightarrow b_2=-1, \quad a_2+b_2-3c_2+1=0$$

$$a_2-1+3+1=0 \Rightarrow a_2=-1$$

$$\pi_2 = \frac{\mu}{\rho U D}, \text{ replace } \pi_2 \rightarrow \pi_2^{-1}, \quad \pi_2 = \frac{\rho U D}{\mu}$$

③ π_3

$$M^0 L^0 T^0 = [L]^{a_3} [L T^{-1}]^{b_3} [M L^{-3}]^{c_3} [T^{-1}]$$

$$= [L]^{a_3+b_3-3c_3} [M]^{c_3} [T]^{-b_3-1}$$

$$c_3=0, \quad a_3+b_3-3c_3=0$$

$$-b_3-1=0 \Rightarrow b_3=-1, \quad a_3-1=0 \Rightarrow a_3=1$$

$$\pi_3 = \frac{N D}{U}$$

$$\frac{F}{\rho U^2 D^2} = f \left[\frac{\rho U D}{\mu}, \frac{N D}{U} \right]$$

Tutorial problem (Solved) The pressure difference (ΔP) in a pipe of diameter D and length l due to turbulent flow depends on velocity V , viscosity μ , density ρ and average height k of roughness projections. Obtain expression for ΔP . $\Delta P = f [D, l, V, \rho, \mu, k]$

Solution $n = 7, m = 3, n - m = 4$

① π_1

$$m^0 L^0 T^0 = [L]^{a_1} [L T^{-1}]^{b_1} [M L^{-3}]^{c_1} M^{d_1} L^{e_1} T^{f_1}$$

$$\begin{bmatrix} \pi_1 - D^{a_1} V^{b_1} \rho^{c_1} \Delta P^{d_1} l^{e_1} \\ \pi_2 - D^{a_2} V^{b_2} \rho^{c_2} l^{e_2} \\ \pi_3 - D^{a_3} V^{b_3} \rho^{c_3} k^{e_3} \\ \pi_4 - D^{a_4} V^{b_4} \rho^{c_4} \mu^{e_4} \end{bmatrix}$$

$$a_1 + b_1 - 3c_1 - 1 = 0 \Rightarrow a_1 - 2 + 3 - 1 = 0$$

$$a_1 = 0$$

$$c_1 + 1 = 0 \Rightarrow c_1 = -1$$

$$-b_1 - 2 = 0 \Rightarrow b_1 = -2$$

$$\pi_1 = D^0 V^{-2} \rho^{-1} \Delta P^1 l^0$$

$$\boxed{\pi_1 = \frac{\Delta P}{\rho V^2}}$$

② π_2

$$m^0 L^0 T^0 = [L]^{a_2} [L T^{-1}]^{b_2} [M L^{-3}]^{c_2} L^{e_2}$$

$$a_2 + b_2 - 3c_2 + 1 = 0$$

$$a_2 + 0 - 3 \times 0 + 1 = 0 \Rightarrow a_2 = -1$$

$$c_2 = 0$$

$$-b_2 = 0$$

$$\boxed{\pi_2 = \frac{l}{D}}$$

$$\boxed{\pi_3 = k/D}$$

③ π_3

Similarly.

④ π_4

$$m^0 L^0 T^0 = [L]^{a_4} [L T^{-1}]^{b_4} [M L^{-3}]^{c_4} M^{d_4} L^{e_4} T^{f_4}$$

$$a_4 + b_4 - 3c_4 - 1 = 0$$

$$a_4 - 1 + 3 - 1 = 0 \Rightarrow a_4 = -1$$

$$c_4 + 1 = 0, -b_4 + 1 = 0$$

$$c_4 = -1, b_4 = 1$$

$$\boxed{\pi_4 = \frac{\mu}{\rho V D}}$$

$$\frac{\Delta P}{\rho V^2} = f \left[\frac{l}{D}, \frac{k}{D}, \frac{\mu}{\rho V D} \right]$$

$$\boxed{\frac{\Delta P}{\rho V^2} = f \left[\frac{l}{D}, \frac{k}{D}, \frac{\rho V D}{\mu} \right]}$$

It can be written as

Reynolds number.

Tutorial problem (Solved)

Show that velocity through a circular orifice is given by

$$V = \sqrt{2gH} \phi \left[\frac{D}{H}, \frac{\mu}{\rho V H} \right]$$

Solution:

$$V = f [H, D, \mu, \rho, g]$$

$$\begin{aligned} n &= 6 \\ m &= 2 \\ n - m &= 4 \end{aligned}$$

$$\pi_1 \rightarrow M^0 L^0 T^0 = H^{a_1} V^{b_1} \rho^{c_1} D$$

$$\pi_2 \rightarrow M^0 L^0 T^0 = H^{a_2} V^{b_2} \rho^{c_2} g$$

$$\pi_3 \rightarrow M^0 L^0 T^0 = H^{a_3} V^{b_3} \rho^{c_3} \mu$$

$$\pi_1 \rightarrow M^0 L^0 T^0 = L^{a_1} [LT^{-1}]^{b_1} [ML^{-3}]^{c_1} L$$

$$\begin{aligned} a_1 + b_1 - 3c_1 + 1 &= 0, & -b_1 &= 0 \\ a_1 + 0 + 0 + 1 &= 0, & c_1 &= 0 \\ a_1 &= -1 \end{aligned}$$

$$\pi_1 = D/H$$

$$\pi_2 \rightarrow M^0 L^0 T^0 = L^{a_2} [LT^{-1}]^{b_2} [ML^{-3}]^{c_2} LT^{-2}$$

$$\begin{aligned} a_2 + b_2 - 3c_2 + 1 &= 0, & -b_2 - 2 &= 0, & c_2 &= 0 \\ a_2 - 2 + 0 + 1 &= 0, & b_2 &= -2 \\ a_2 &= 1 \end{aligned}$$

$$\pi_2 = \frac{H g}{V^2}$$

$$\pi_3 \rightarrow M^0 L^0 T^0 = L^{a_3} [LT^{-1}]^{b_3} [ML^{-3}]^{c_3} ML^{-1}T^{-1}$$

$$\begin{aligned} c_3 + 1 &= 0, & -b_3 - 1 &= 0, & a_3 + b_3 - 3c_3 - 1 &= 0 \\ c_3 &= -1, & b_3 &= -1, & a_3 - 1 + 3 - 1 &= 0 \Rightarrow a_3 = -1 \end{aligned}$$

$$\pi_3 = \frac{\mu}{\rho V H}$$

replace $\pi_2 = \frac{1}{\sqrt{\pi_2}} = \frac{V}{\sqrt{2gH}}$ replace by $\frac{V}{\sqrt{2gH}}$

$$V = \sqrt{2gH} \phi \left[\frac{D}{H}, \frac{\mu}{\rho V H} \right]$$

Tutorial problem (Solved)

A pipe of diameter 1.5m is required to transport oil of specific gravity 0.9 and $\mu = 3 \times 10^{-2}$ poise at a rate of 3000 litres/s. Tests were conducted on 15cm diameter pipe using water. Find velocity and rate of flow in model.
 $\mu_{\text{water}} = 0.01$ poise.

Solution

For similar pipe flows Reynolds number must be same

$$\boxed{Re_m = Re_p}$$

$$Q_p = 3000 \text{ litres/s} = 3 \text{ m}^3/\text{s}$$
$$V_p = \frac{3}{\pi/4 \times (1.5)^2} = 1.697 \text{ m/s}$$

$$\left(\frac{\rho V D}{\mu} \right)_m = \left(\frac{\rho V D}{\mu} \right)_p$$

$$\frac{1000 \times V \times 0.15}{0.01} = \frac{0.9 \times 1000 \times 1.697 \times 1.5}{3 \times 10^{-2}}$$
$$V_m = \frac{0.9 \times 1.697 \times 1.5}{3 \times 0.15}$$

$$V_m = 5.091 \text{ m/s}$$

$$Q_m = \pi/4 \times (0.15)^2 \times 5.091$$
$$= 0.008996 \text{ m}^3/\text{s}$$
$$= \underline{\underline{89.96 \text{ litres/s}}}$$

Tutorial problem (Solved)

A rectangular pier 1.2m wide $\times$ 4m long is to be built in a river whose average depth is 3m. To determine the discharge characteristics, a small scale model made to scale of 1:20 was tested in a laboratory and the following data were recorded. velocity = 0.6 m/s, Height of standing wave = 3.5cm, force acting on model = 4N.

make calculations for Speed, Force, height of standing wave and Coefficient of drag resistance for the prototype. Assume that flows in model and prototype are insensitive to changes in Reynolds number.

Solution

Flow involves free surface flows hence gravity force predominate and Froude number will be used for dynamic similarity.

$$\frac{V_m}{\sqrt{L_m g}} = \frac{V_p}{\sqrt{L_p g}} \Rightarrow \frac{V_m}{V_p} = \sqrt{\frac{L_m}{L_p}}$$

$$V_p = 0.6 \times \sqrt{20} = 2.683 \text{ m/s}$$

$$\text{Acceleration scale ratio} = \frac{a_m}{a_p} = \frac{V_m / T_m}{V_p / T_p} = \frac{V_m}{V_p} \times \frac{T_p}{T_m} = \frac{V_m}{V_p} \times \frac{L_p}{L_m} \times \frac{V_m}{V_p}$$

$$= \left(\frac{V_m}{V_p}\right)^2 \times 20$$

$$\frac{T_m}{T_p} = \frac{L_m / V_m}{L_p / V_p} = \frac{L_m}{L_p} \times \frac{V_p}{V_m}$$

$$\frac{T_p}{T_m} = \frac{L_p}{L_m} \times \frac{V_m}{V_p}$$

$$\frac{a_m}{a_p} = 1$$

$$\text{Scale factor for force ratio} = \frac{F_m}{F_p} = \frac{\rho_m a_m}{\rho_p a_p} = \frac{\rho_m}{\rho_p} = \frac{L_m L_m^2}{L_p L_p^2} \quad (\text{if } \rho_m = \rho_p)$$

$$F_p = F_m \frac{\rho_p}{\rho_m} \cdot \left(\frac{L_p}{L_m}\right)^3 = 4 \times (20)^3 = 32000 \text{ N}$$

From geometric similarity $\frac{h_m}{h_p} = \frac{L_m}{L_p} \Rightarrow h_p = 3.5 \times 20 = 70 \text{ cm}$
 height of standing wave in prototype

$$\text{Drag coefficient} = \frac{F}{\frac{1}{2} \rho A V^2} = \frac{32000}{\frac{1}{2} \times 1000 \times (1.2 \times 3) \times (2.68)^2} = 2.475 \quad (\text{same for model and prototype})$$